Shortest Path Problems

- Single source
  - I want to find the shortest paths from a given node to all other nodes
- Point to point
  - I want to find the shortest path from A to B
  - Typically done by stopping the single source problem early

https://www.youtube.com/watch?v=T_m27bhVQQQ

Weighted vs Unweighted Costs

- Unweighted path cost is the number of edges in the path
- Weighted path cost is the sum of the costs of the edges in the path

Required Data

- Representation of the graph
- Array to keep track of the path cost
- Array to keep track of the previous vertex
  - needed to construct the actual path
- Array to keep track of whether we've found a path to that vertex
- A queue

What Goes in the Queue?

- The vertex we just reached
- The vertex we came from
- The total cost of the path to this vertex
  - The number of edges back to the starting vertex

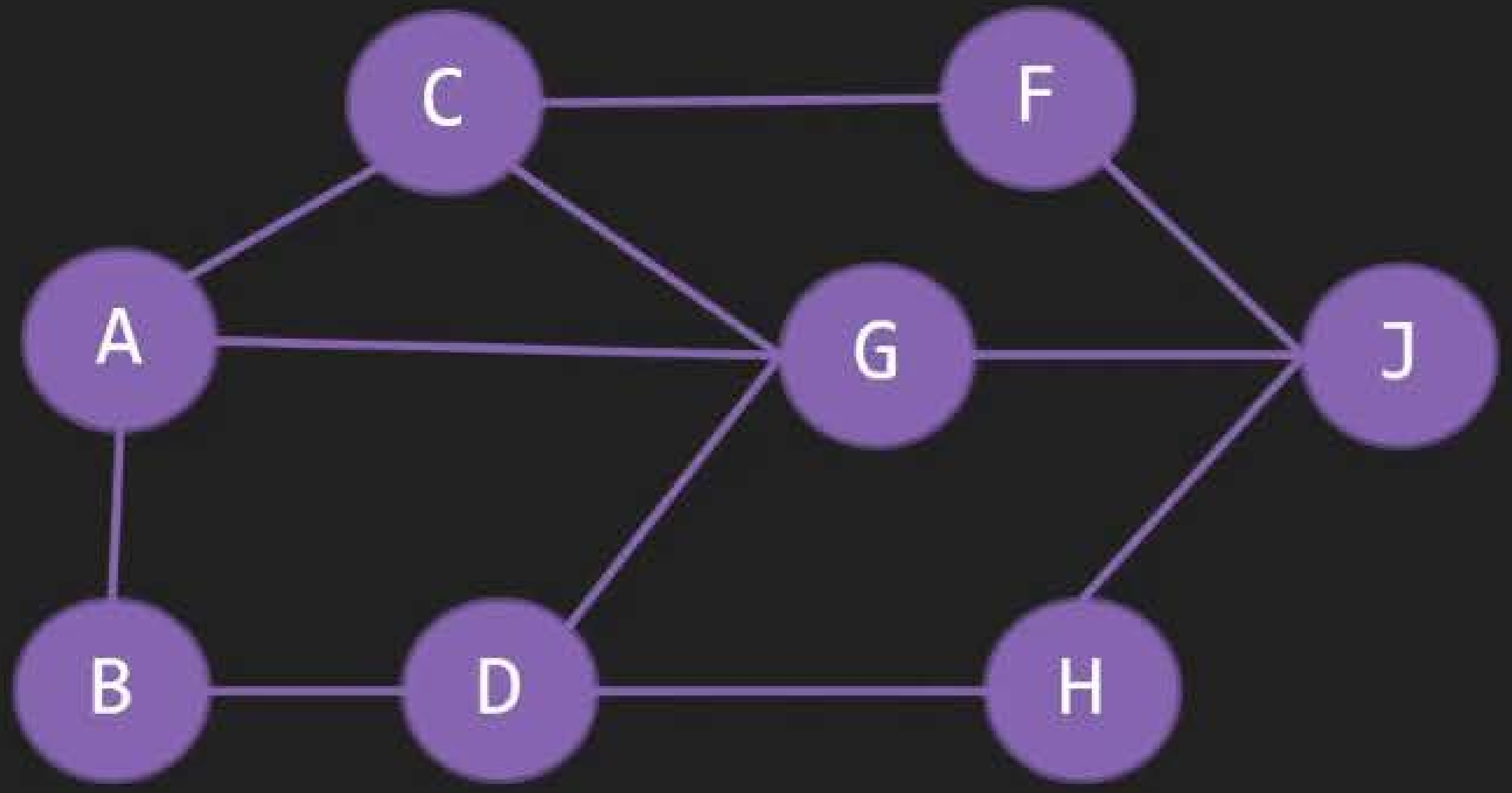
Process

Add all vertices directly connected to the source to the queue

While the queue is not empty AND there are vertices we don't have paths to

- Take the first item off the queue
- If there is not yet a path to that vertex
  - Update the arrays
  - For each edge out of the vertex
    - If there is no path to the ending vertex yet
      - Add that vertex information to the queue

Handle First Item from Queue



Queue

From	D	J
To	H	H
Cost	3	3

Path and Cost Arrays

A	-	-	✓
B	A	1	✓
C	A	1	✓
D	G	2	✓
F	C	2	✓
G	A	1	✓
H	D	3	✓
J	G	2	✓

Find Path from A to H

Reverse the sequence to get the path

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H D G A

A G D H  -- with a cost of 3
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A	-	-
B	A	1
C	A	1
D	G	2
F	C	2
G	A	1
H	D	3
J	G	2